

Mixed Reality Radio Visualization

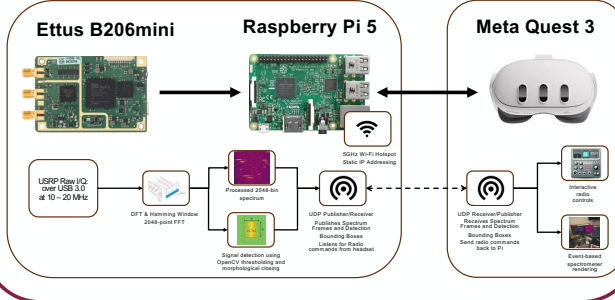
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Motivation

- **Problem:** Traditional spectrum analyzers are tethered to bulky 2D screen, making it difficult to maintain situational awareness while locating physical radio frequency (RF) sources
- **Solution:** Mixed Reality (MR) spectral data onto the physical world, enabling users to see the signals in their working environment
- **Use Cases:**
 - RF "Fox Hunt" Transmitter Locating
 - Identifying RF leakage, ...

System Architecture



Signal Detection

- **Method:** OpenCV based Thresholding with Morphological Closing
- **Process:**
 - Apply binary threshold to normalized waterfall
 - Merge active FFT bins using 9 x 9 kernel
 - Extract bounding boxes with bounding function
- **Coordinate Mapping:** Translating circular buffer indices into (xBin, yTime) coordinates for Unity.

Objectives

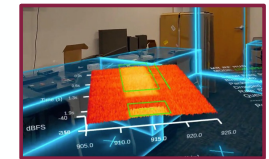
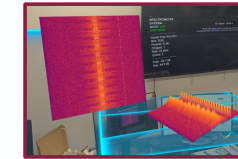
- **Real-Time Visualization:** Achieve <XXms latency
- **Edge Processing:** Offload heavy computation to edge
- **Headset Control:** Tune SDR from MR environment
- **Portability:** Operate without being tethered
- **Signal Insight:** Provide insight to user (e.g. detection)

I/Q Acquisition Pipeline

- USRP → fc32 Buffer → Hamming Window → FFTW Execution → UDP Package
- **Buffer Management:**
 - MTU Optimized Slices: Captured larger data chunks to maximize USB 3.0 throughput.
 - Bounded Data Queue: Enqueued FFT slices into a bounded queue to manage backpressure.
- **Pre-allocation Strategy:** Used aligned buffers to avoid memory copying during computations

3D Waterfall Rendering

- Data Flow: int8 dB Array → Unity Texture Row → Shader Vertex Offset → 3D MR Overlay
- **GPU-Accelerated Scrolling:** A Texture2D acts as a ring buffer. A custom shader is used to map the color and height of the spectrogram
- **Color Mapping:** Pre-computed Inferno LUT
- **Bounding Box Overlays:** Detection results are mapped and rendered as on top of the scrolling waterfall



RF Concepts

- **I/Q Sampling:**
$$s(t) = I(t) + jQ(t) = A(t)e^{j\phi(t)}$$
- **Discrete Fourier Transform (DFT):**

The transformation used to move signal from the time domain to the frequency domain:

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-j\frac{2\pi}{N}nk}$$
- **Hamming Window:**

Applied to $w[n]$ before the FFT to reduce spectral leakage by smoothing the edges of the block:

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right)$$
- **Magnitude Scaling:** $P_{dB} = 20 \cdot \log_{10}(|X[k]| + \epsilon)$

Networking

- Raspberry Pi as Wi-Fi Hotspot
- **Protocol:** UDP chosen for low latency
- **UDP Packages:**
 - Spectrum Frame: Pkg #, Frequency, dB Array
 - Detection Frame: Bounding Box Coordinates
- **Ports:**
 - 5005: Pi → Quest (Spectrum & Detection Frames)
 - 5006: Quest → Pi (Radio Tuning Commands)
 - 5007: **UPDATE**
 - 5008: Pi → Quest (Operation Status)

Future Work

- **ML Integration:** Moving from OpenCV thresholding to YOLOv8 for automatic signal classification (e.g., Identifying Wi-Fi vs. Bluetooth).
- **Frequency Hopping:** Implementing the hopping logic to track agile signals.
- **Multi-SDR Support:** Synchronizing multiple SDRs for TDOA (Time Difference of Arrival) localization.



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